

A Project Report on

CAMPUSPAL : SMART CAMPUS ASSISTANT

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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2025-2026



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CERTIFICATE

This is to certify that the project entitled “CampusPal : Generative AI-Driven Institutional Information Assistant” is a bonafide work of **Vaibhav Bura(22106067), Yash Desai(22106005), Vivek Dalvi(22106108), Pratik Dhas(22106063)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering in Computer Science & Engineering (Artificial Intelligence & Machine Learning)**.

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Abstract

Educational institutions maintain a vast amount of academic and administrative information across websites, PDF documents, and various digital platforms. Although this information is accessible online, it is often scattered and difficult to navigate, making it challenging for students, parents, and faculty to obtain accurate and relevant information efficiently. Users are typically required to browse multiple pages and documents, which is time-consuming and often leads to confusion. Traditional search systems rely on keyword-based approaches and lack contextual understanding, resulting in incomplete or irrelevant outputs. Moreover, general-purpose AI chatbots may generate inaccurate or misleading responses due to the absence of domain-specific knowledge.

To address these challenges, this project presents **CampusPal**, an AI-powered institutional assistant developed for A. P. Shah Institute of Technology (APSIT). The system is designed to provide instant, accurate, and context-aware responses to queries related to admissions, fee structures, placements, faculty details, and academic policies. CampusPal utilizes a Retrieval-Augmented Generation (RAG) approach, which combines semantic retrieval with controlled language generation to ensure that responses are grounded in verified institutional data.

The system is built using a modular architecture consisting of data collection, preprocessing, embedding generation, vector storage, and response generation. Institutional data is extracted from official sources such as websites and documents, cleaned, and segmented into meaningful chunks. These chunks are converted into vector embeddings and stored in a FAISS-based vector database to enable efficient semantic search. When a user submits a query, the system retrieves the most relevant information and generates a response using a controlled language model.

CampusPal improves accessibility to institutional information, reduces dependency on manual navigation, and enhances user experience by delivering reliable and context-aware responses. The system provides a scalable and efficient solution that can be extended to other educational institutions for improved information retrieval and user support.

Keywords : Artificial Intelligence, Natural Language Processing, Retrieval-Augmented Generation, Semantic Search, Chatbot, Institutional Information System, Vector Database, FAISS, Context-Aware Response, Information Retrieval

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ABBREVIATION

AI	<i>Artificial Intelligence</i>
ML	<i>Machine Learning</i>
NLP	<i>Natural Language Processing</i>
LLM	<i>Large Language Model</i>
RAG	<i>Retrieval-Augmented Generation</i>
FAISS	<i>Facebook AI Similarity Search</i>
API	<i>Application Programming Interface</i>
UI	<i>User Interface</i>
DPR	<i>Dense Passage Retrieval</i>
IR	<i>Information Retrieval</i>
QA	<i>Question Answering</i>

Chapter 1

Introduction

1. Introduction

In today's digital era, educational institutions store a large volume of academic and administrative information across websites, PDF documents, and online portals. This includes details about admissions, fee structures, faculty, placements, and academic policies. Although this information is available online, it is often scattered across multiple sources, making it difficult for users to access relevant information efficiently.

Students, parents, and faculty frequently spend significant time navigating through different web pages to find specific details. Traditional search systems rely on keyword-based matching and lack contextual understanding, which often results in irrelevant or incomplete information [1]. Moreover, general-purpose AI chatbots may generate inaccurate responses due to the absence of domain-specific knowledge, which is a critical issue in academic environments where accuracy is essential [2].

To address these challenges, this project introduces **CampusPal**, an AI-powered institutional assistant developed for A. P. Shah Institute of Technology (APSIT). The system is designed to provide instant, accurate, and context-aware responses to queries related to institutional information such as admissions, fees, placements, and faculty details.

CampusPal uses a Retrieval-Augmented Generation (RAG) approach, which combines semantic search with controlled response generation [3][4]. Instead of relying solely on pre-trained knowledge, the system retrieves relevant information from a structured institutional database and uses it to generate responses. This ensures that the answers are factual, reliable, and aligned with official data.

The overall working of the system involves collecting institutional data, processing it into structured formats, storing it in a vector database, and retrieving relevant information based on user queries. The system then generates responses using a controlled language model, ensuring accuracy and contextual relevance.

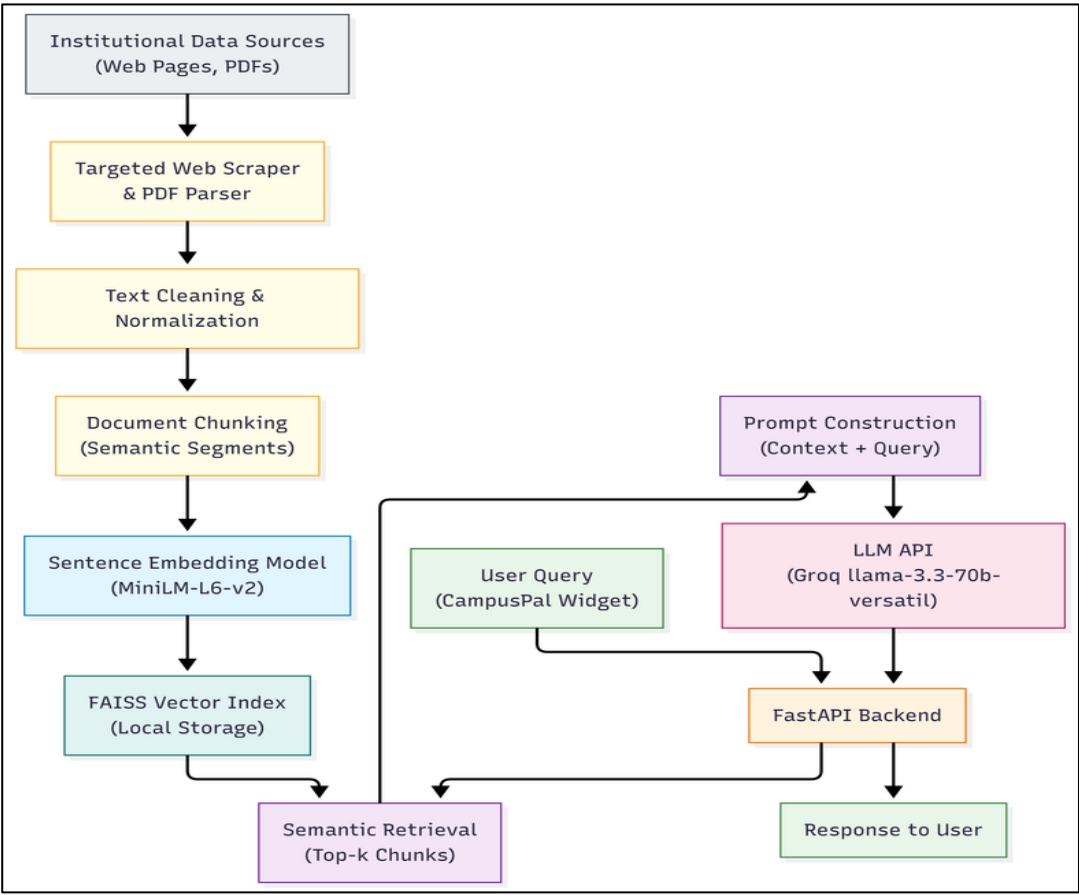


Figure 1 : High Level Overview Of CampusPal System

Figure 1 illustrates the overall workflow of the CampusPal system. It shows how user queries are processed through the backend, relevant information is retrieved from the vector database, and responses are generated using a controlled language model.

Chapter 2

Literature Survey/ Existing System

2. Survey/ Existing System

The development of intelligent conversational systems has gained significant attention in recent years due to the rapid advancement of Artificial Intelligence and Natural Language Processing. This chapter reviews existing research in conversational AI, educational chatbots, large language models, retrieval-based systems, and vector search techniques. The analysis highlights the strengths and limitations of current approaches and identifies the research gap addressed by the proposed system.

2.1 Conversational AI Systems

Conversational Artificial Intelligence (AI) focuses on enabling machines to interact with humans using natural language. Early conversational systems were primarily rule-based and relied on predefined scripts and pattern-matching techniques. These systems lacked flexibility and were unable to handle variations in user queries, making them unsuitable for complex interactions.

With the introduction of machine learning techniques, conversational systems evolved to incorporate statistical methods for intent recognition and response generation. Neural network-based models further improved performance by enabling systems to learn from

large datasets and adapt to different user inputs [5]. These models allowed for better understanding of user intent and more natural responses.

The development of deep learning architectures, particularly transformer-based models, marked a significant breakthrough in conversational AI [6]. The transformer architecture introduced attention mechanisms that allow models to capture long-range dependencies in text, improving contextual understanding. This advancement enabled the development of more sophisticated conversational systems capable of maintaining context across multiple interactions.

Despite these improvements, conversational AI systems still face challenges in maintaining long-term context, ensuring accuracy, and handling domain-specific queries. These limitations highlight the need for systems that combine conversational capabilities with reliable information retrieval mechanisms.

2.2 Chatbots in Education

Chatbots have been increasingly adopted in educational environments to automate responses to frequently asked questions and improve access to information [7][8]. These systems are used for providing details about admissions, course structures, examination schedules, and institutional policies.

Rule-based chatbots are commonly used in educational settings due to their simplicity and ease of implementation [9]. However, they are limited in their ability to handle complex queries and require manual updates to incorporate new information. Machine learning-based chatbots improve flexibility but depend heavily on training data and require periodic retraining.

Research indicates that educational chatbots must be domain-specific to ensure accuracy and relevance. Generic chatbots often provide incorrect or vague responses due to lack of access to institutional data. Furthermore, many existing systems fail to support multi-turn conversations, limiting their effectiveness in real-world scenarios.

Another important challenge in educational chatbots is maintaining consistency and reliability of information. Since academic information is frequently updated, systems must ensure that responses are based on the latest data. This has led to the exploration of retrieval-based approaches that dynamically fetch information from updated sources.

2.3 Large Language Models

Large Language Models (LLMs) such as transformer-based architectures have significantly advanced the field of natural language processing [10]. Models like BERT and GPT are based on transformer architectures and are trained on large-scale datasets.

These models have demonstrated strong performance in various tasks, including question answering, text summarization, and conversational response generation [11]. The ability of LLMs to understand context and generate coherent responses makes them suitable for chatbot applications [12].

However, LLMs have a critical limitation known as hallucination, where the model generates information that is not grounded in real data [2]. This issue is particularly problematic in academic applications, where incorrect information can lead to serious consequences.

Additionally, LLMs rely on pre-trained knowledge and may not have access to updated or domain-specific information. This makes them less reliable for applications requiring accurate and real-time data. These limitations have led researchers to explore hybrid approaches that combine LLMs with external knowledge sources.

2.4 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is a hybrid approach that combines information retrieval with language generation to improve response accuracy [3][4]. Instead of relying solely on the model's internal knowledge, RAG retrieves relevant documents from a knowledge base and uses them as context for generating responses as shown in Figure 2.

The retrieval component uses dense vector representations to identify relevant documents based on semantic similarity. Dense Passage Retrieval (DPR) is a commonly used technique for this purpose [13]. The retrieved documents are then passed to a language model, which generates responses grounded in the provided context.

This approach offers several advantages over traditional generative models. By incorporating external knowledge, RAG reduces hallucination and improves factual accuracy [2]. It also allows the system to access updated information without retraining the model.

Research has shown that RAG significantly improves performance in knowledge-intensive tasks such as question answering and information retrieval. This makes it particularly suitable for applications like academic chatbots, where accuracy and reliability are critical.

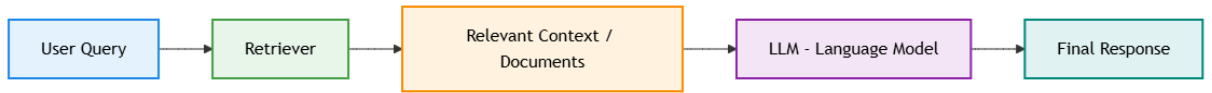


Figure 2 : RAG Workflow Diagram

2.5 Vector Search and FAISS

Efficient retrieval of relevant information is a key component of RAG-based systems [14]. Vector search techniques enable semantic matching by converting text into numerical representations known as embeddings.

FAISS (Facebook AI Similarity Search) is a widely used library for efficient similarity search in large datasets. It allows fast retrieval of nearest neighbor vectors, making it suitable for real-time applications.

By using vector embeddings, systems can perform semantic search instead of relying on keyword matching. This improves retrieval accuracy and ensures that relevant information is selected for response generation. FAISS also supports large-scale datasets, making it scalable for real-world applications.

2.6 Supporting Technologies in NLP Systems

Several foundational technologies contribute to the development of modern conversational systems. Transfer learning techniques allow models to leverage pre-trained knowledge and adapt to specific tasks [15]. Sequence-to-sequence models enable end-to-end learning for text generation tasks [16][17].

Frameworks such as transformer libraries provide tools for implementing state-of-the-art NLP models. Benchmark datasets like SQuAD are used to evaluate the performance of question-answering systems.

These technologies collectively enable the development of advanced conversational systems that can handle complex queries and provide accurate responses [18].

2.7 Hybrid Retrieval Techniques in Conversational Systems

Hybrid retrieval techniques combine multiple retrieval approaches to improve the accuracy and relevance of information retrieval systems [19]. Traditional systems rely either on keyword-based search or semantic search independently, which can limit performance in

complex query scenarios. Keyword-based methods ensure precise matching, while dense retrieval techniques capture semantic meaning through vector representations.

Recent research suggests that combining these approaches improves both precision and recall in information retrieval tasks. Dense retrieval methods such as Dense Passage Retrieval (DPR) enable semantic understanding of queries, while traditional methods ensure that exact matches are not overlooked [13]. This hybrid strategy is particularly useful in conversational systems where user queries can vary in structure and intent.

In academic chatbot applications, hybrid retrieval helps in handling both general and specific queries. For example, queries related to admission procedures may require semantic interpretation, whereas queries about exact fee amounts require precise data retrieval. By integrating hybrid retrieval techniques, systems can ensure more accurate and contextually relevant responses.

Thus, hybrid retrieval enhances the robustness of conversational systems by balancing semantic understanding with exact data matching, making it highly suitable for institutional information systems.

2.8 Context Awareness and Multi-Turn Dialogue Systems

Context awareness is a critical aspect of modern conversational AI systems, especially in multi-turn dialogue scenarios [20]. Users often ask follow-up questions that depend on previous interactions, and systems must retain context to provide meaningful responses.

Traditional chatbot systems lack the ability to maintain conversational context, leading to disjointed interactions. Recent advancements in transformer-based models have improved context handling through attention mechanisms that capture relationships between different parts of a conversation [6]. These models enable better understanding of user intent across multiple queries.

In educational environments, context-aware systems are essential. For instance, a user may first ask about admission eligibility and then inquire about fees. A context-aware system can link both queries and provide coherent responses without requiring the user to repeat information.

Additionally, integrating conversational history with retrieval mechanisms further improves system performance. By considering previous interactions during retrieval, the system can generate responses that are both contextually relevant and factually accurate. This approach enhances user experience and reduces redundancy in conversations.

Table 1 illustrates the summary of existing research works related to conversational AI, retrieval techniques, and chatbot systems, highlighting their approaches, datasets, and identified gaps.

Table 1: Literature Survey

YEAR	DOMAIN	ALGORITHM / APPROACH	DATASET	GAP
[3] 2020	Retrieval-Augmented Generation	RAG (Retriever + Generator)	Wikipedia, Open QA	High computational cost
[13] 2020	Dense Passage Retrieval	Dense Vector Retrieval	Open-domain QA datasets	Requires large training data
[7] 2019	Sentence-BERT	Sentence Embeddings	STS, NLI datasets	Needs fine-tuning for domain use
[23] 2021	FAISS Similarity Search	Vector Search	Large-scale embeddings	Memory and indexing challenges
[6] 2017	Transformer Model	Attention Mechanism	WMT dataset	High computation requirement
[8] 2022	AI Chatbots in Education	Rule-based + ML	Educational datasets	Limited contextual understanding
[27] 2022	Neural IR Survey	Neural Retrieval	IR benchmarks	Scalability issues

Chapter 3

Limitation of Existing System

3. Limitation of Existing System

Existing information retrieval systems and chatbot solutions used in educational institutions exhibit several limitations that affect their efficiency, accuracy, and usability. Traditional institutional websites primarily rely on static content and keyword-based search mechanisms. These systems require users to manually navigate through multiple pages and documents to locate relevant information. Such an approach is time-consuming and often leads to confusion, especially for new users who are unfamiliar with the structure of the website.

Rule-based chatbot systems, which are commonly used in educational environments, operate on predefined patterns and responses [9]. While they are effective for handling simple and repetitive queries, they lack the flexibility to process complex or dynamic user inputs. Any new query pattern requires manual updates to the system, making maintenance difficult and limiting scalability.

Machine learning-based chatbots improve upon rule-based systems by learning from data; however, they depend heavily on the quality and quantity of training datasets. These systems require frequent retraining to adapt to new information, which is not always practical in rapidly changing academic environments. Additionally, they may struggle with unseen

queries, leading to inaccurate or incomplete responses.

General-purpose AI chatbots, powered by large language models, provide more natural and conversational interactions. However, they suffer from a critical limitation known as hallucination, where the model generates responses that are not grounded in factual data [2]. In the context of educational institutions, such inaccuracies can mislead users and reduce trust in the system.

Another major limitation of existing systems is the lack of domain-specific knowledge integration. Many chatbots are not directly connected to institutional data sources, resulting in generic or outdated responses. Furthermore, most systems lack context awareness and fail to handle multi-turn conversations effectively. Users are often required to repeat their queries, which reduces the overall user experience.

3.1 Research Gap

From the analysis of existing systems, it is evident that there is a need for an intelligent solution that combines accuracy, scalability, and contextual understanding. Current approaches either rely on static rule-based systems or unrestricted generative models, both of which have inherent limitations.

There is a clear gap in the development of systems that integrate domain-specific institutional data with advanced retrieval and generation techniques. Existing solutions do not effectively utilize semantic search mechanisms alongside controlled language generation to ensure factual accuracy.

Additionally, limited focus has been given to combining context-aware conversational capabilities with reliable data retrieval in educational environments. Most systems fail to provide consistent and contextually relevant responses across multiple interactions.

To address these gaps, there is a need for a system that incorporates Retrieval-Augmented Generation (RAG), enabling it to retrieve verified information from institutional sources and generate accurate responses [3][4]. Such a system should also support context-aware interactions, scalable architecture, and real-time performance.

The proposed system, CampusPal, is designed to bridge this gap by integrating semantic retrieval, vector-based search, and controlled response generation. This ensures that users receive accurate, context-aware, and reliable information tailored to the institutional domain.

Chapter 4

Problem Statement and Objective

4. Problem Statement and Objective

Educational institutions store academic and administrative information across multiple digital platforms such as websites, documents, and online portals. However, this information is often fragmented and difficult to navigate, making it challenging for users to retrieve accurate and relevant details efficiently. Traditional keyword-based search systems lack contextual understanding, while general-purpose AI chatbots may generate inaccurate responses due to the absence of domain-specific knowledge. Hence, there is a need for an intelligent system that can provide accurate, context-aware, and reliable information retrieval in real time.

4.1 Objectives

- To develop an AI-powered institutional assistant, CampusPal, for efficient retrieval of academic and administrative information.
- To provide a conversational interface that enables users to interact using natural language and obtain accurate responses.
- To implement a Retrieval-Augmented Generation (RAG) framework that combines semantic search with controlled response generation to ensure factual accuracy.
- To reduce hallucination in AI-generated responses by grounding outputs in verified institutional data.
- To ensure fast response time and efficient information retrieval through optimized system design.
- To design a scalable and modular system architecture for handling institutional queries.
- To enhance user experience by providing reliable, context-aware, and easy-to-access information.

4.2 Scope

- Institution-Specific Information Retrieval :

The system is designed to provide accurate and relevant information specifically related to A. P. Shah Institute of Technology (APSIT), including admissions, fees, placements, faculty details, and academic policies, ensuring domain-specific responses.

- Real-Time Query Processing :

The system is capable of handling user queries in real time, ensuring quick and efficient response generation through optimized retrieval and backend processing mechanisms.

- Reduction of Hallucination in Responses :

By using Retrieval-Augmented Generation and controlled prompt design, the system ensures that responses are generated based on verified data, reducing incorrect or misleading outputs.

Chapter 5

Proposed System

5. Proposed System

The proposed system, CampusPal, is an AI-powered institutional assistant designed to provide accurate, context-aware, and real-time responses to queries related to A. P. Shah Institute of Technology (APSIT). The system employs advanced Natural Language Processing techniques integrated with a Retrieval-Augmented Generation (RAG) framework to ensure that responses are grounded in verified institutional data rather than relying solely on generative capabilities.

The architecture of the system follows a modular and scalable design consisting of data acquisition, preprocessing, embedding generation, vector storage, retrieval, and response generation components [21]. Unlike traditional chatbot systems that depend on static responses or generic language models, the proposed system dynamically retrieves relevant information from a structured knowledge base and generates responses based on contextual understanding.

The overall system operates in a pipeline where user queries are processed, relevant context is retrieved from a vector database, and responses are generated using a controlled language model. This approach ensures high accuracy, contextual relevance, and reduced hallucination [2].

5.1 System Framework and Architecture

The proposed system follows a layered architecture consisting of three primary layers: Data Layer, Application Layer, and Presentation Layer. The Data Layer is responsible for collecting and preprocessing institutional data from various sources such as websites and documents. The Application Layer performs semantic retrieval and response generation using embeddings and language models. The Presentation Layer provides an interactive interface for user interaction.

The architecture ensures seamless integration between components, enabling efficient data flow from input to output.

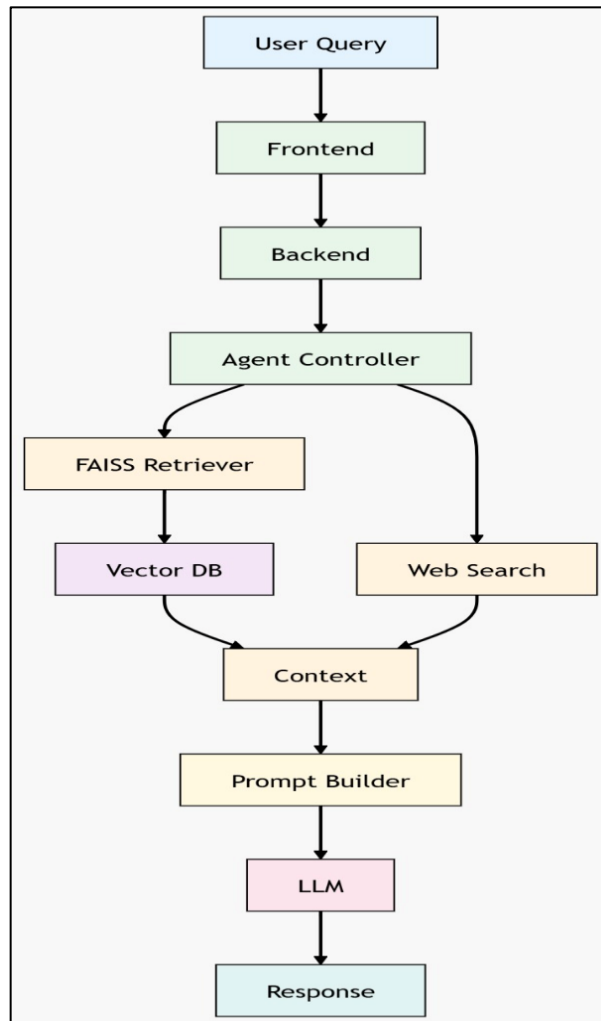


Figure 3 : Proposed System Architecture

Figure 3 illustrates the overall architecture of the system, showing the flow of user queries through the backend, retrieval module, and response generation component. It highlights how the system integrates vector search with a language model to generate accurate responses.

5.2 Data Acquisition and Preprocessing

The system collects institutional data from multiple sources, including official websites, PDF documents, and structured digital content. A custom scraping mechanism is utilized to extract meaningful textual data while filtering out irrelevant elements such as navigation components and formatting tags.

The collected data undergoes preprocessing to improve quality and usability. This process includes cleaning, normalization, removal of redundant content, and segmentation into smaller chunks [22]. Chunking is performed to maintain contextual integrity while enabling efficient retrieval. Each chunk is designed to preserve semantic meaning and is optimized for embedding generation.

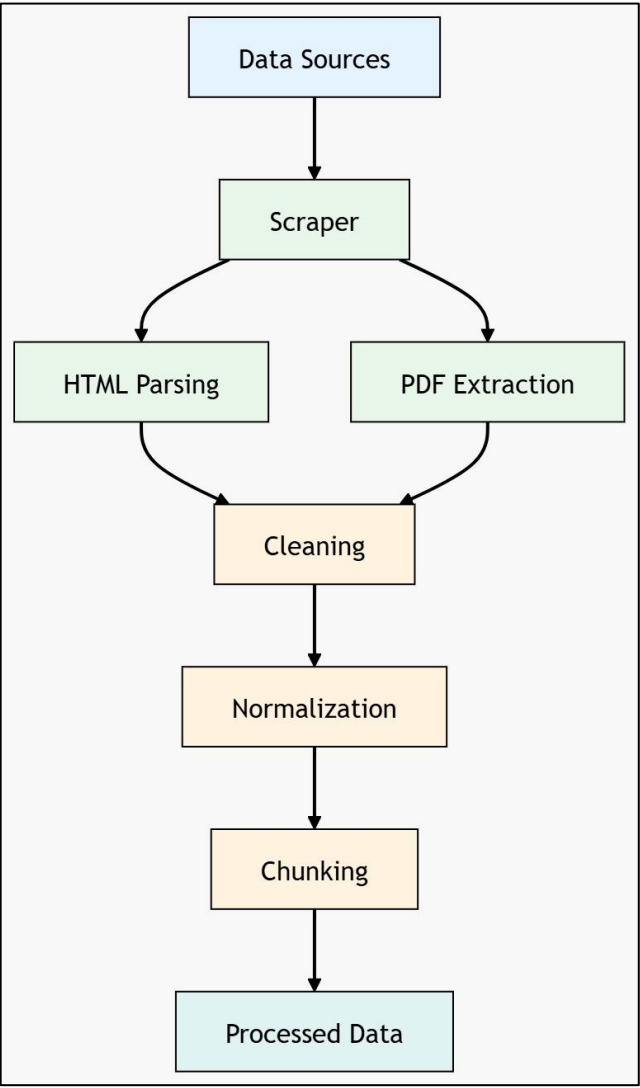


Figure 4 : Data Processing Pipeline

Figure 4 represents the data pipeline, illustrating how raw institutional data is processed into structured textual segments suitable for embedding and storage.

5.3 Embedding and Vector Storage

The processed textual data is converted into vector embeddings using a pre-trained sentence transformer model, specifically the all-MiniLM-L6-v2 model. This model generates embeddings of dimension 384, capturing the semantic meaning of text in a high-dimensional space.

These embeddings are stored in a FAISS (Facebook AI Similarity Search) vector database, which enables efficient similarity-based retrieval. FAISS supports fast nearest-neighbor search, allowing the system to handle large datasets while maintaining real-time performance [23][24].

The use of vector embeddings enables the system to perform semantic search rather than relying on exact keyword matching, significantly improving retrieval accuracy [22].

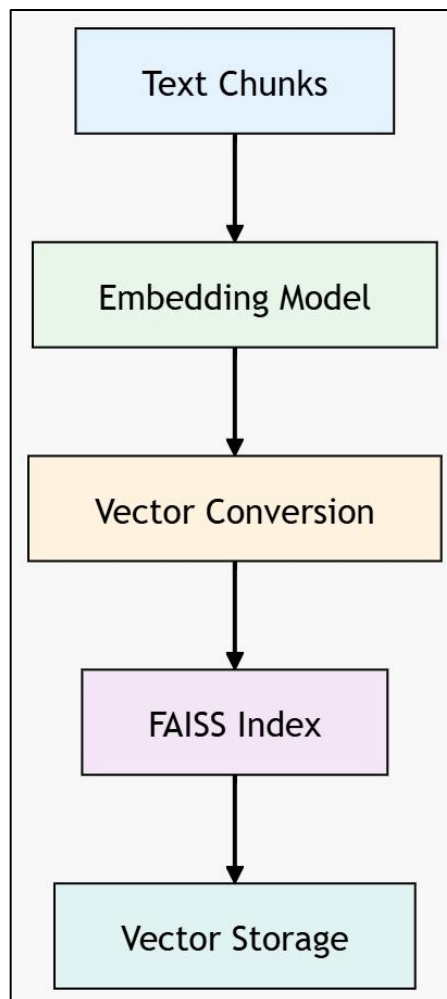


Figure 5 : Embedding and Storage Flow

Figure 5 illustrates the transformation of textual data into embeddings and their storage in the FAISS vector database.

5.4 Query Processing and Retrieval

When a user submits a query, the system processes it by converting the query into a vector embedding using the same embedding model. This ensures consistency between stored data and incoming queries.

The system performs similarity search in the FAISS database to retrieve the top $k = 5$ most relevant text chunks [25][23][24]. The use of top-k retrieval enhances contextual richness and improves the quality of generated responses. The retrieved chunks form the contextual input required for response generation.

This semantic retrieval mechanism enables the system to understand user intent even when queries are phrased differently, ensuring flexibility and robustness.

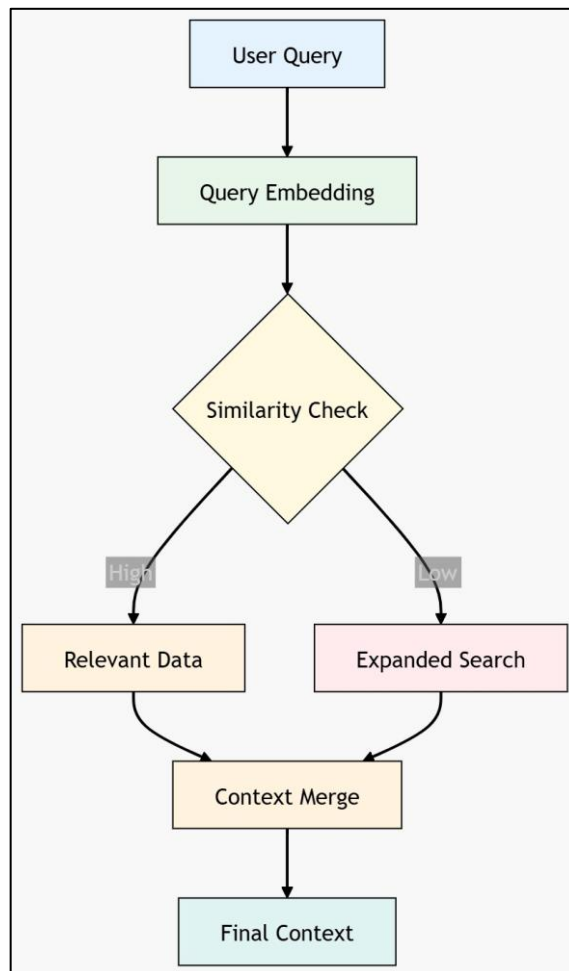


Figure 6 : Query Processing Flow

Figure 6 demonstrates how user queries are processed and matched with relevant data in the vector database.

5.5 Response Generation using Retrieval-Augmented Generation

The system utilizes Retrieval-Augmented Generation (RAG) to generate responses. The retrieved context and user query are combined to form a structured prompt, which is then processed by a language model [3][4][19].

The model generates responses strictly based on the provided context, ensuring factual accuracy and reliability. A low temperature value of 0.1 is used to reduce randomness and produce deterministic outputs. Additionally, strict prompt design is employed to ensure that the model does not generate information outside the retrieved context.

This approach significantly reduces hallucination and ensures that responses remain aligned with institutional data.

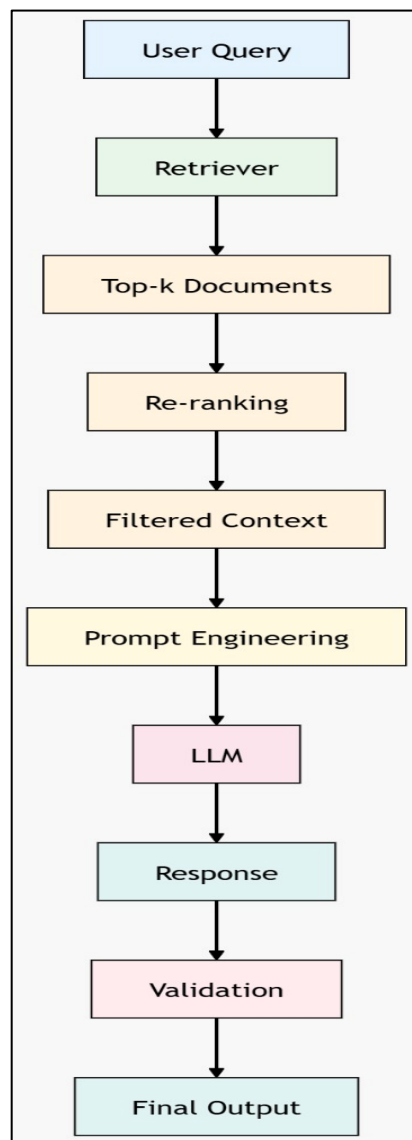


Figure 7 : RAG Workflow

Figure 7 illustrates the integration of retrieval and generation components, showing how contextual information is used to generate accurate responses.

5.6 Backend and Frontend Design

The backend of the system is implemented using FastAPI, which handles API requests, manages data flow, and ensures efficient communication between system components. FastAPI supports asynchronous processing, enabling the system to handle multiple requests with low latency.

The frontend is developed using React.js, providing a responsive and user-friendly interface. Users can interact with the system through natural language queries, and responses are displayed in a conversational format. The interface enhances accessibility and improves user experience.



Figure 8: CampusPal Integration with Institutional Website (Prototype Demonstration)

Figure 8 illustrates a prototype demonstration of integrating the CampusPal chatbot into an institutional website interface. The chatbot is displayed as an interactive widget that enables users to access information directly from the homepage.

This integration is developed for demonstration purposes to showcase how the system can be deployed in a real-world environment. The current implementation of the project focuses on building the core AI-powered chatbot system, while website integration represents a potential extension for future deployment.

The prototype highlights the scalability and adaptability of the system across institutional platforms.

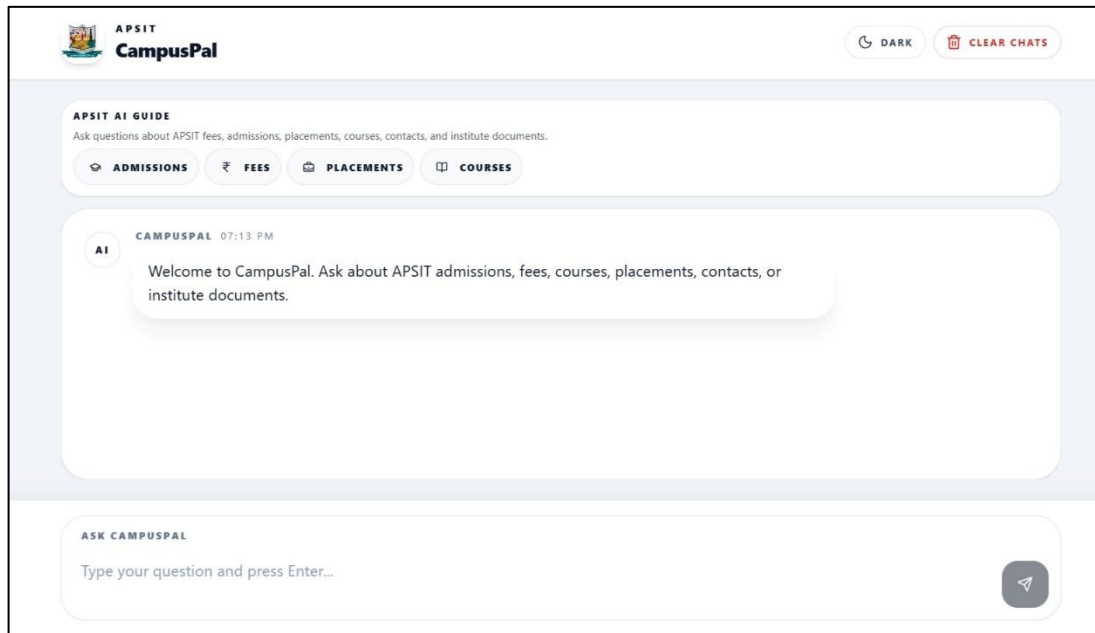


Figure 9: CampusPal Main Chat Interface

Figure 9 shows the main user interface of the CampusPal chatbot. The interface provides a simple and interactive environment where users can ask queries related to institutional information.

Predefined options such as admissions, fees, placements, and courses are provided to guide users and improve accessibility. The interface is designed to be user-friendly and supports real-time interaction between the user and the system.

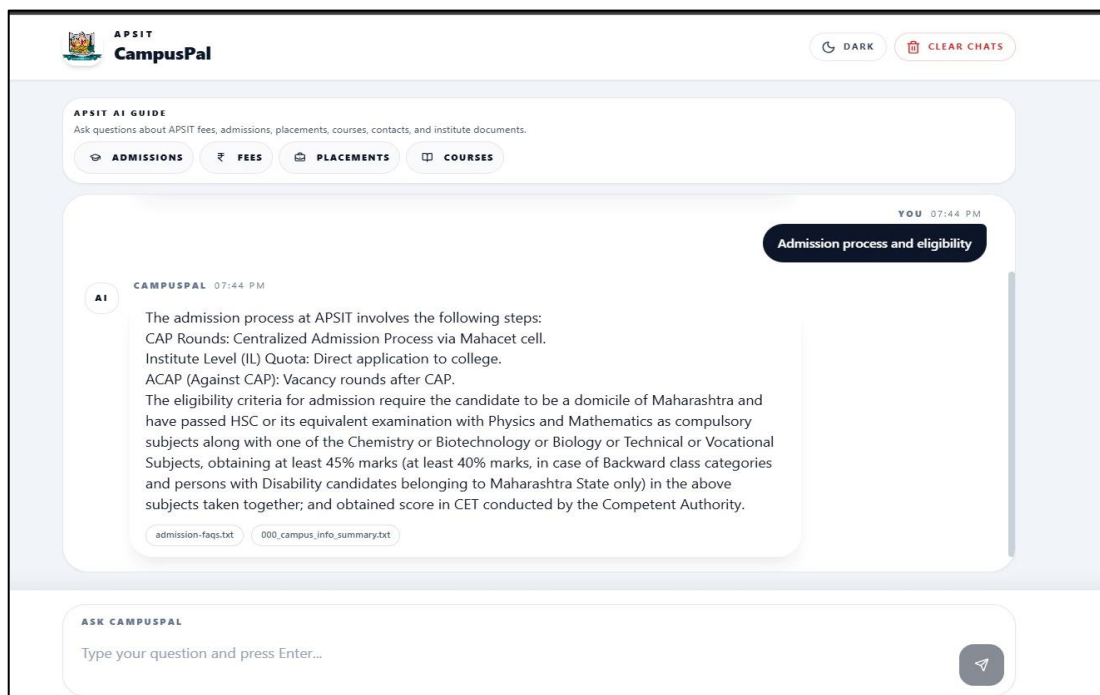


Figure 10: Admission and Entrance Examination Query Responses

Figure 10 demonstrates the system's ability to handle queries related to admission processes and entrance examinations. The chatbot retrieves relevant institutional data and provides structured and accurate responses.

The responses include detailed steps of the admission process and accepted entrance examinations, indicating effective semantic retrieval and context-based answer generation.

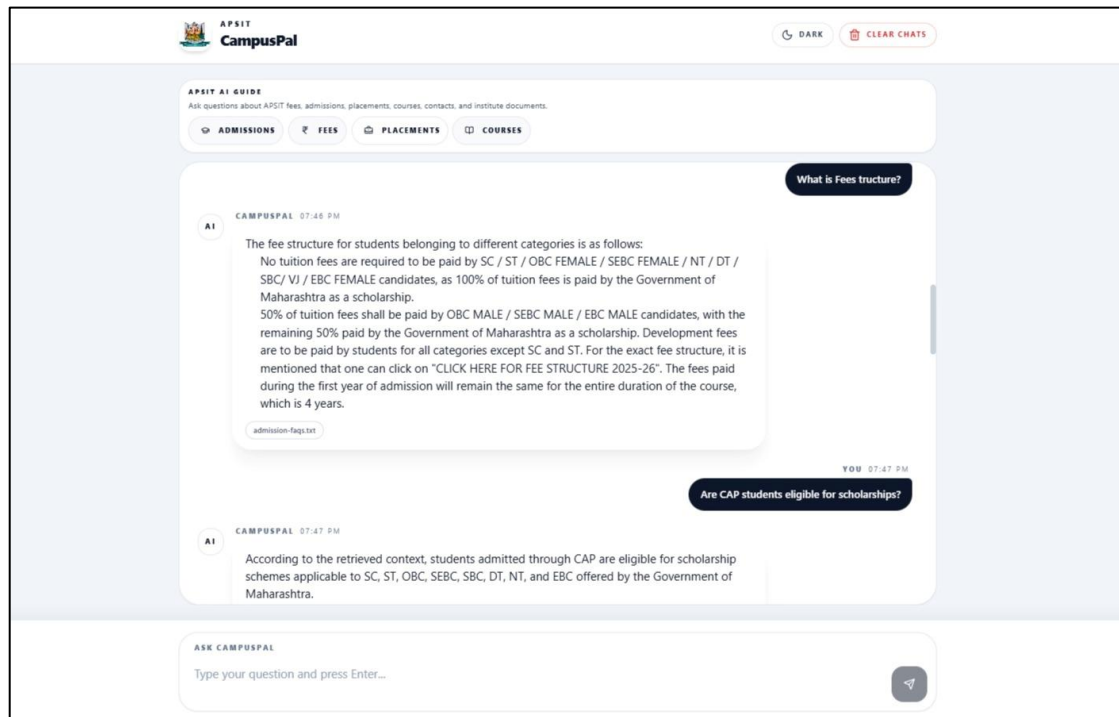


Figure 11: Fee Structure and Scholarship Query Responses

Figure 11 presents responses generated by the system for queries related to fee structure and scholarship eligibility. The chatbot accurately provides category-wise fee details and scholarship information based on retrieved data.

The system demonstrates its capability to handle structured financial queries and generate precise, context-aware responses, which are critical for institutional information systems.

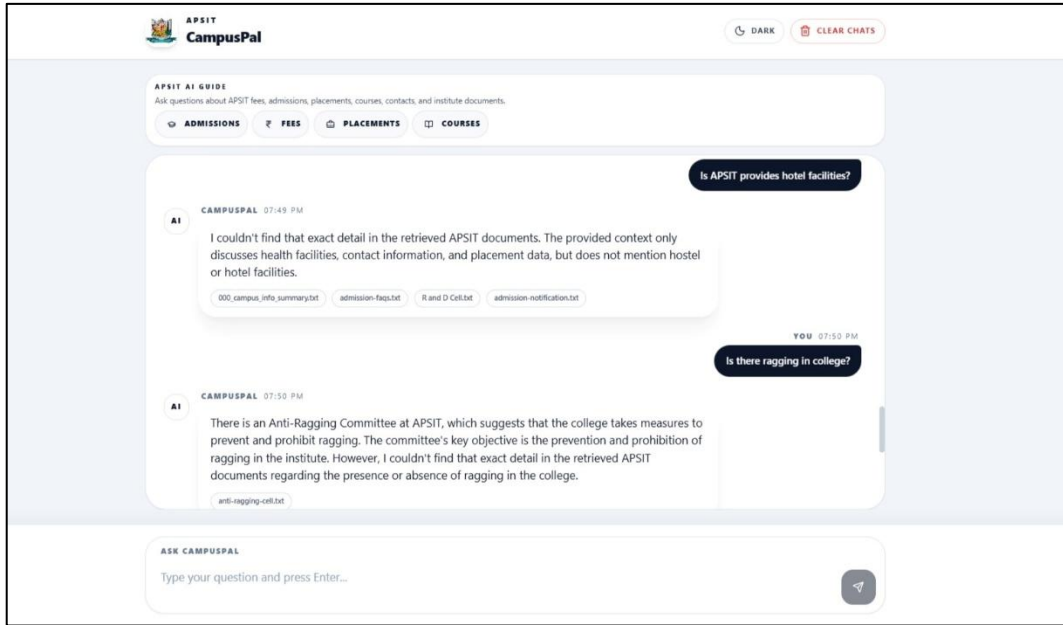


Figure 12: Handling of Missing or Partial Information Queries

Figure 12 illustrates how the system handles queries where complete or exact information is not available in the dataset. Instead of generating incorrect or misleading responses, the chatbot provides safe fallback answers by indicating the absence of specific information. This behavior demonstrates the system's ability to reduce hallucination and maintain reliability by grounding responses only in available data.

The presented interface and response examples demonstrate the practical implementation of the CampusPal system. The results confirm that the system is capable of delivering accurate, context-aware, and reliable responses for institutional queries, while also safely handling incomplete or unavailable information.

5.7 Methodology

The methodology of the proposed system follows a structured pipeline beginning with data collection from institutional sources, followed by preprocessing and segmentation of textual data into smaller chunks [26]. These chunks are converted into vector embeddings and stored in a FAISS database [23][24].

Upon receiving a user query, the system processes the query, retrieves relevant contextual information using similarity search, and generates a response using a Retrieval-Augmented Generation framework. The generated response is then delivered to the user through the interface.

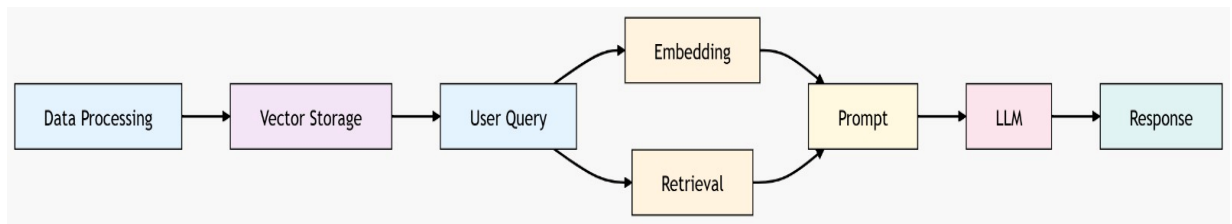


Figure 13 : End-to-End Workflow

Figure 13 represents the complete flow of the system from data collection to response generation.

5.8 Advantages of Proposed System

The proposed system offers several advantages, including improved accuracy through semantic retrieval, reduction of hallucination using Retrieval-Augmented Generation, and enhanced accessibility of institutional information [27]. The system supports context-aware responses, enabling better understanding of user queries. Additionally, the modular architecture ensures scalability and adaptability, allowing the system to be extended to other institutions with minimal modifications.

Chapter 6

Experimental Setup

6. Experimental Setup

The experimental setup of the proposed CampusPal system includes the software and hardware components required for development, deployment, and evaluation. The system utilizes modern AI frameworks, web technologies, and suitable hardware resources to ensure efficient processing and real-time response generation.

Table 2 illustrates the software components used in the development of the proposed system along with their respective roles in implementation.

Table 2: Software Configuration

COMPONENT	DESCRIPTION
Programming Language	Python
Backend Framework	FastAPI
Frontend	React.js (Vite)
AI Framework	LangChain
Language Model	LLaMA (via Groq API)
Vector Database	FAISS

Web Scraping Tools	BeautifulSoup, Pandas
Development Environment	Visual Studio Code
Version Control	Github

Table 3 illustrates the hardware requirements necessary for executing the proposed system efficiently.

Table 3: Hardware Configuration

COMPONENT	SPECIFICATION
Processor	Intel i5 or higher
RAM	Minimum 8 GB
Storage	At least 10 GB free space
Operating System	Windows / Linux
Internet	Required for API access

The above configuration supports efficient data processing, semantic retrieval, and real-time interaction for the CampusPal system.

6.1 Dataset Description

The dataset used in the system consists of domain-specific institutional data extracted from official APSIT sources such as web pages and PDF documents. The collected data includes structured and unstructured information related to various academic and administrative aspects.

The extracted data was processed and converted into a structured textual format suitable for semantic retrieval. The preprocessing pipeline involved cleaning the data removing irrelevant content, and segmenting it into smaller chunks to preserve contextual meaning.

Each text segment was converted into vector embeddings and stored in a FAISS-based vector database, enabling efficient similarity-based retrieval during query processing. The system accepts user queries in natural language format, which may include questions related to admissions, fees, placements, faculty details, and academic policies.

Table 4 illustrates the characteristics of the dataset used in the system, including data source, type, processing methods, and storage mechanism.

Table 4: Dataset Description

PARAMETER	DESCRIPTION
Data Source	APSIT Website and Documents
Data Type	Text (Structured + Unstructured)
Processing	Cleaning, Chunking, Embedding
Storage	FAISS Vector Database

6.2 Performance Evaluation Parameters (Validation)

The performance of the proposed system was evaluated through practical testing using a set of institutional queries. The evaluation focused on how the system behaves during real-time interaction rather than theoretical measurement of model performance.

A collection of queries related to admissions, fees, placements, faculty details, and academic policies was used to test the system. For each query, the retrieved context and generated response were manually observed and compared with the expected institutional information.

During testing, the system's ability to retrieve relevant data, generate appropriate responses, and maintain consistency was examined. The response time was also observed to ensure that the system performs efficiently in real-time scenarios.

Instead of using predefined evaluation formulas, the assessment was carried out by observing system output across multiple queries and verifying whether the responses were correct, relevant, and consistent.

Table 5 illustrates the key evaluation aspects used to assess system performance during testing along with observed outcomes.

Table 5: Evaluation Aspects

TEST ASPECT	OBSERVATION DURING TESTING
Query Coverage	System handled most institutional queries correctly
Retrieval Behavior	Relevant context retrieved in majority of cases

Response Quality	Responses aligned with institutional data
Consistency	Similar queries produced stable responses
Response Time	Responses generated within short time duration

6.3 System Implementation Environment

The proposed system was implemented using a combination of modern web technologies and AI frameworks to ensure efficient performance and scalability.

The backend of the system was developed using Python and FastAPI, which handles API requests, query processing, and interaction with the vector database. The frontend was implemented using React.js, providing an interactive and user-friendly interface for query input and response display.

The system uses FAISS for vector storage and similarity search, along with a sentence transformer model for embedding generation. The response generation component is powered by a language model integrated through an API, enabling fast and context-aware responses.

Table 6 illustrates the key technologies used in the proposed CampusPal system along with their respective roles. These components collectively enable efficient data processing, semantic retrieval, and context-aware response generation, forming the core functionality of the system.

Table 6: Technologies Used in Proposed System

COMPONENT	ROLE IN SYSTEM
Python	Core programming language for backend logic
FastAPI	Handles API requests and backend processing
React.js	Provides user interface for interaction
FAISS	Enables fast semantic similarity search
LangChain	Manages RAG pipeline and workflow
LLaMA (via Groq API)	Generates context-aware responses

Chapter 7

Results and Discussion

7. Results and Discussion

The proposed system, CampusPal, was evaluated based on its ability to provide accurate, context-aware, and real-time responses to institutional queries. The evaluation primarily focused on response correctness, relevance of retrieved information, and system behavior under different types of queries.

The system was tested using a variety of queries related to admissions, fee structures, placements, faculty information, and academic policies. These queries were selected to reflect practical usage scenarios within an academic environment. The observations indicate that the system is capable of retrieving relevant information and generating responses that are aligned with institutional data.

7.1 Observations from System Testing

During the testing phase, the system showed stable and consistent performance across different categories of queries. It was observed that the retrieval mechanism was effective in identifying relevant information from the vector database, especially for structured queries such as fee details and admission procedures.

The use of semantic search enabled the system to interpret user queries even when they were not phrased using exact keywords. In most cases, the system was able to return appropriate responses without requiring query reformulation. The integration of Retrieval-Augmented Generation ensured that responses were based on retrieved context rather than generated assumptions.

The response time was observed to be low and suitable for real-time interaction [28]. The use of efficient vector search through FAISS contributed significantly to reducing latency during query processing.

7.2 Response Analysis

The responses generated by the system were analyzed based on correctness, contextual relevance, and clarity. It was observed that the system consistently produced responses derived from the retrieved data, which ensured reliability and reduced the occurrence of incorrect outputs.

In cases where relevant data was available in the dataset, the responses were complete and aligned with user intent. For queries with partial or limited data, the system generated concise responses without introducing incorrect information. This behavior indicates that the system prioritizes accuracy over unnecessary generation.

Table 7: Detailed Query Results

TEST QUERY	CAMPUSPAL ANSWER (SUMMARY)	RETRIEVAL HIT	FAITHFUL	ANALYSIS
Admission process and eligibility	Correct admission details	Yes	Yes	Success
Which entrance exam accepts?	MHT-CET or JEE	Yes	Yes	Success (follow-up handled)
What is amount of the fees?	Irrelevant “Design registration” info	No	N/A	Retrieval failure
Are CAP students eligible for scholarships?	Yes, CAP students eligible	Yes	Yes	Success

Fee for EBC female students?	No tuition fees	Yes	Yes	Success
And for male?	50% tuition fee	Yes	Yes	Success (follow-up handled)
Is hostel facility available?	No hostel available	Yes	Yes	Success
Is there ragging in college?	Anti-ragging committee info	Yes	Yes	Success
How to visit the campus?	Generic made-up steps	No	No	Hallucination
Who is principal of college?	Irrelevant “conflict of interest” context	No	Yes (to wrong context)	Retrieval failure

Table 7 presents a detailed evaluation of the chatbot using representative institutional queries. The analysis indicates that the system successfully retrieved relevant context and generated faithful responses for the majority of queries. Out of 10 test queries, 7 were handled successfully, where both retrieval and response generation were accurate.

However, in 3 cases, the system failed due to retrieval limitations. Two queries resulted in irrelevant responses because incorrect context was retrieved, while one query led to a hallucinated response where no relevant information was found. These observations highlight that system performance is highly dependent on the quality and coverage of the underlying dataset.

Table 8 illustrates the system behavior across different types of queries, highlighting response accuracy and contextual relevance.

Table 8 : Sample Query Results

QUERY TYPE	SYSTEM RESPONSE	OBSERVATION
Admission Query	Accurate	Relevant and complete

Fee Structure	Accurate	Precise information
Placement Info	Relevant	Context-based response
Faculty Details	Accurate	Retrieved from dataset
General Query	Limited	Depends on available data

7.3 System Performance Analysis

The performance of the system was analyzed qualitatively with respect to retrieval efficiency and response generation behavior. The use of semantic embeddings enabled better matching between user queries and stored data compared to traditional keyword-based approaches.

The system retrieves the top 5 relevant chunks ($k = 5$), which provides sufficient contextual information for generating meaningful responses [29]. This improves the quality of answers, especially for queries that require multiple pieces of information.

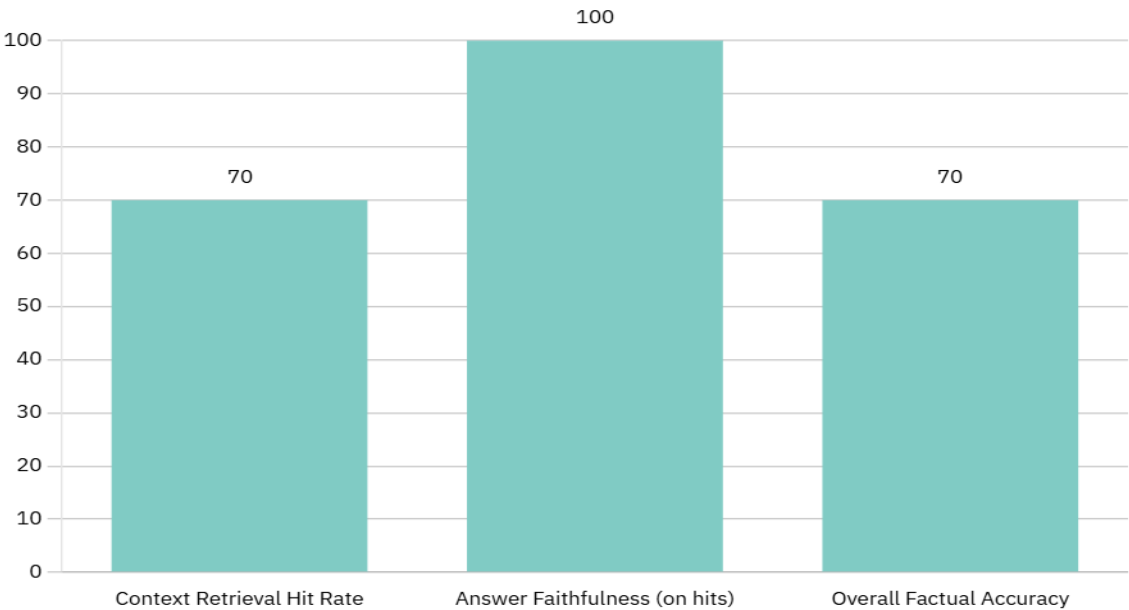
The use of a low temperature setting in the language model ensured deterministic and consistent responses. It was observed that repeated queries produced similar outputs, indicating stability in response generation.

Table 9: Performance Metrics

METRIC	TOTAL QUERIES	SUCCESSFUL	ACCURACY
Context Retrieval Hit Rate	10	7	70%
Answer Faithfulness	7 (from hits)	7	100%
Overall Factual Accuracy	10	7	70%

Table 9 summarizes the overall performance of the system based on the test queries. The context retrieval hit rate was observed to be 70%, indicating that relevant information was successfully retrieved in most cases. Among the correctly retrieved cases, the system achieved 100% answer faithfulness, demonstrating that the generated responses were fully aligned with the retrieved context.

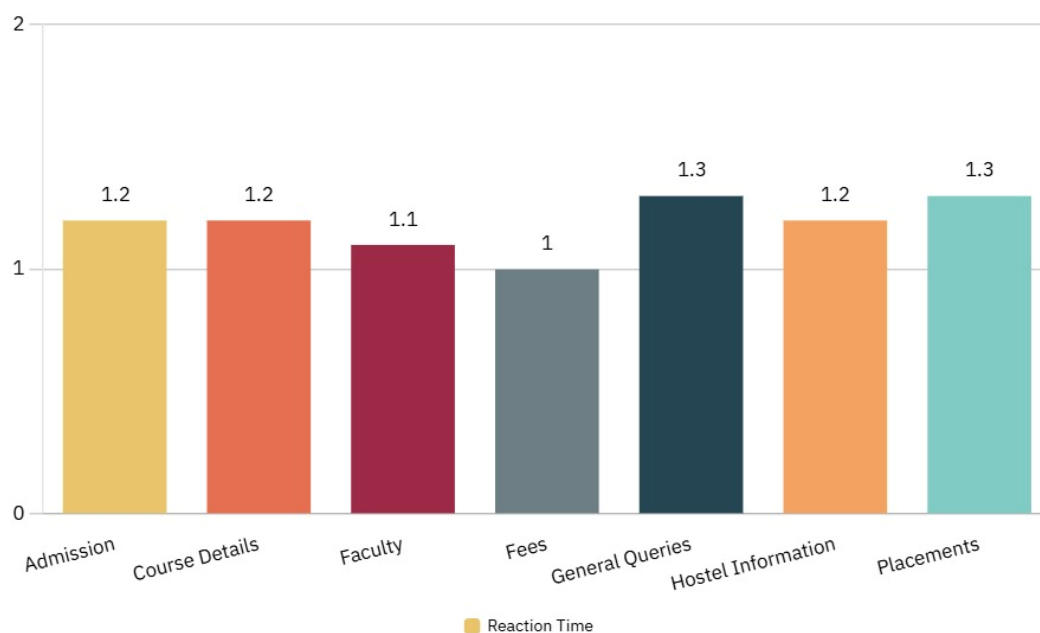
The overall factual accuracy of the system was recorded as 70%, reflecting the dependency of response correctness on retrieval quality. These results confirm that the proposed system performs effectively when relevant context is available, while highlighting the need for improving data coverage to handle edge cases.



Graph 1: CampusPal Performance Metrics

Graph 1 illustrates the performance of the system across key evaluation metrics. The graph shows that answer faithfulness achieved the highest score of 100%, indicating that the system generates accurate responses when correct context is retrieved. The context retrieval hit rate and overall factual accuracy were both observed to be 70%, reflecting the dependency of system performance on effective retrieval.

The results emphasize that the primary limitation of the system lies in retrieving the correct information rather than generating responses. Improving the dataset quality and retrieval mechanism can further enhance overall system performance.



Graph 2: Response Time Analysis

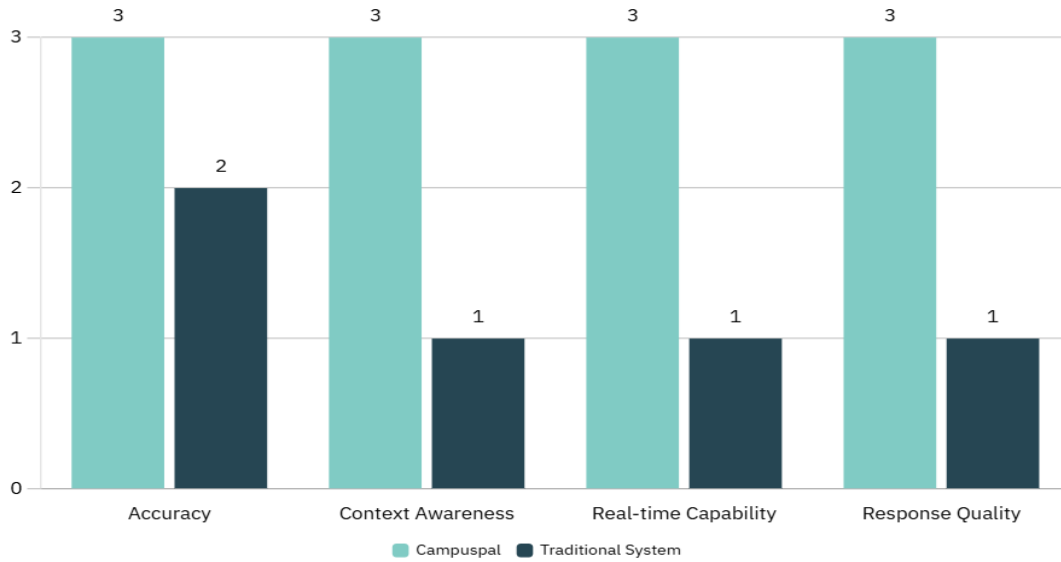
Graph 2 represents the response time observed during system testing. It indicates that the system is capable of generating responses within a short duration, making it suitable for real-time applications.

7.4 Comparison with Traditional Systems

The proposed system was compared with traditional keyword-based search systems to evaluate improvements in information retrieval and response generation. Table 10 illustrates the comparison between traditional keyword-based systems and the proposed CampusPal system in terms of performance and capabilities.

Table 10 : Comparison with Traditional Systems

FEATURE	TRADITIONAL SYSTEM	CAMPUSPAL
Search Method	Keyword-based	Semantic Search
Accuracy	Moderate	High
Context Awareness	Limited	Improved
Response Format	Static	Conversational
Real-Time Response	Limited	Supported



Graph 3: System Comparison

The comparison shown in Graph 3 indicates that the proposed system performs better in terms of contextual understanding and response relevance. The use of semantic retrieval allows the system to interpret user intent more effectively than keyword-based approaches [30].

7.5 Discussion

The results demonstrate that the proposed system provides a reliable solution for institutional information retrieval. By combining semantic search with Retrieval-Augmented Generation, the system is able to produce responses that are both accurate and context-aware.

The integration of FAISS enables efficient retrieval of relevant data, while the controlled response generation mechanism reduces the likelihood of incorrect outputs. The system maintains consistency in responses and performs effectively across different types of queries.

However, the performance of the system is dependent on the quality and completeness of the dataset. In cases where information is not available, the system provides limited responses, which highlights the importance of maintaining a comprehensive and updated dataset.

Overall, the system demonstrates practical applicability in academic environments and provides a significant improvement over traditional information retrieval methods.

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

8.1 Conclusion

The proposed system, CampusPal, successfully addresses the challenges associated with accessing institutional information in an efficient and structured manner. Educational data is often distributed across multiple sources, making it difficult for users to retrieve accurate and relevant information. The proposed system overcomes these limitations by integrating semantic retrieval with controlled response generation.

The system employs a Retrieval-Augmented Generation (RAG) framework, which combines vector-based retrieval with language model-based response generation [3][4]. By utilizing FAISS for similarity search and sentence transformer-based embeddings, the system enables efficient semantic matching of user queries with institutional data. The use of top-k retrieval ($k = 5$) ensures that sufficient contextual information is provided for generating accurate responses.

To enhance reliability, the system incorporates controlled generation techniques, including low temperature settings (0.1) and strict prompt design, which significantly reduce hallucinated outputs. This ensures that the responses are grounded in verified institutional data and remain consistent across different queries.

The system demonstrates the ability to handle a wide range of queries related to admissions, fee structures, placements, faculty information, and academic policies. It provides context-aware and concise responses while maintaining real-time performance. The modular architecture of the system further ensures scalability and flexibility, allowing it to be extended to other educational institutions.

Overall, CampusPal provides an effective and reliable solution for institutional information retrieval. It enhances accessibility, reduces dependency on manual assistance, and demonstrates the practical application of AI in academic environments.

8.2 Future Work

The proposed system can be further enhanced by incorporating additional features to improve functionality and user experience. One possible extension is the implementation of multilingual support, which would allow users to interact with the system in multiple languages, thereby increasing accessibility for a diverse user base.

Another potential improvement is the integration of voice-based interaction using speech recognition and text-to-speech technologies. This would enable users to communicate with the system through voice commands, making the system more interactive and user-friendly. The system can also be integrated with official institutional databases to enable real-time data updates. This would ensure that the information provided by the system remains accurate and up-to-date without relying solely on static data sources.

Further enhancements can be made in context handling by improving the system's ability to manage long and multi-turn conversations. This would allow the system to provide more coherent and contextually consistent responses over extended interactions.

Additionally, the development of a dedicated mobile application can improve accessibility and provide users with a more convenient platform for interaction. Personalization features can also be incorporated to deliver responses tailored to individual users based on their preferences and interaction history.

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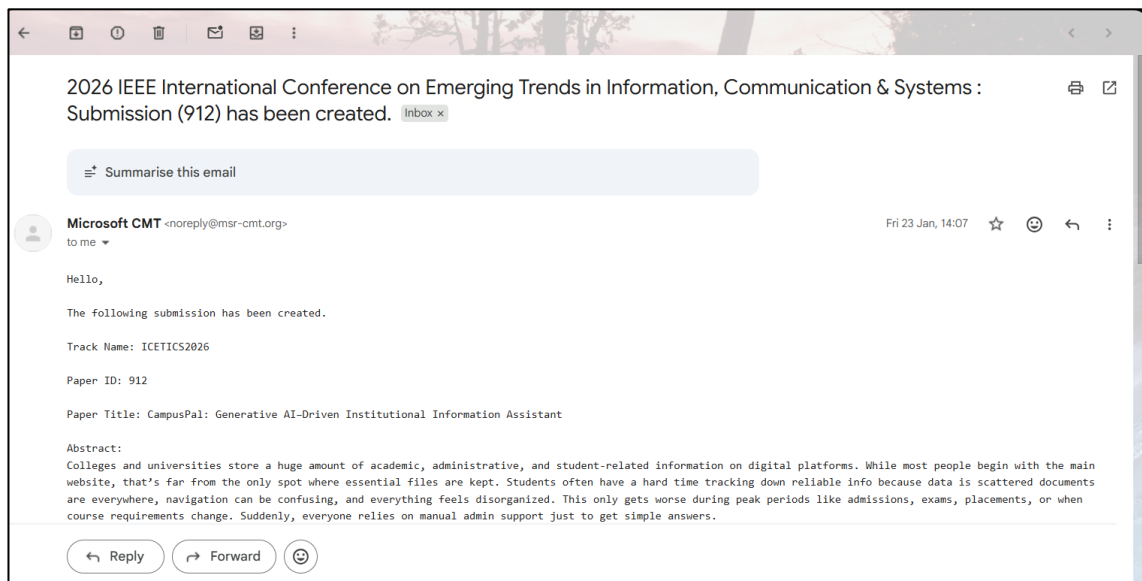
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